

Progress test 1

1. Vector differential calculus

- (a) Calculate of the area of a triangle, the vertices of which are given by [20%]

$$(1, 1, 2), (3, -1, 0), (-1, 0, -2).$$

- (b) Prove the following inequality if $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$. [30%]

$$\|\mathbf{x} + \mathbf{y} + \mathbf{z}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| + \|\mathbf{z}\|.$$

- (c) Sketch $z = -x^2/4 - y^2 + 4$ for $z \geq 0$, and find the tangent plane at $(x, y) = (1, 1)$. [30%]

- (d) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and $g : \mathbb{R}^3 \rightarrow \mathbb{R}$. Prove the following vector differential identity: [20%]

$$\nabla^2(fg) = f\nabla^2g + g\nabla^2f + 2(\nabla f \cdot \nabla g).$$

$$(a) \quad \|a \times b\| = \|a\| \|b\| \sin \theta$$

$$\therefore A = \frac{1}{2} \|a\| \|b\| \sin \theta = \frac{1}{2} \|a \times b\|$$

$$= \frac{1}{2} \|(1, 1, 2) - (-1, 0, -2)\| \times \|(3, -1, 0) - (-1, 0, -2)\|$$

$$= \frac{1}{2} \|(2, 1, 4) \times (4, -1, 2)\|$$

$$= \frac{1}{2} \|(2+4, 16-4, -2-4)\|$$

$$= \frac{1}{2} \|(6, 12, -6)\|$$

$$= \frac{1}{2} \times 6\sqrt{6} = 3\sqrt{6}$$

$$b) (\|x+y+z\|)^2 = (x+y+z)^2$$

$$= x^2 + y^2 + z^2 + 2(x \cdot y + x \cdot z + y \cdot z)$$

$$(\|x\| + \|y\| + \|z\|)^2 = \|x\|^2 + \|y\|^2 + \|z\|^2 + 2(\|x\|\|y\| + \|x\|\|z\| + \|y\|\|z\|)$$

$$= x^2 + y^2 + z^2 + 2(\|x\|\|y\| + \|x\|\|z\| + \|y\|\|z\|)$$

$$\therefore (\|x\| + \|y\| + \|z\|)^2 - (\|x+y+z\|)^2$$

$$= 2(\|x\|\|y\| - x \cdot y) + 2(\|x\|\|z\| - x \cdot z) + 2(\|y\|\|z\| - y \cdot z)$$

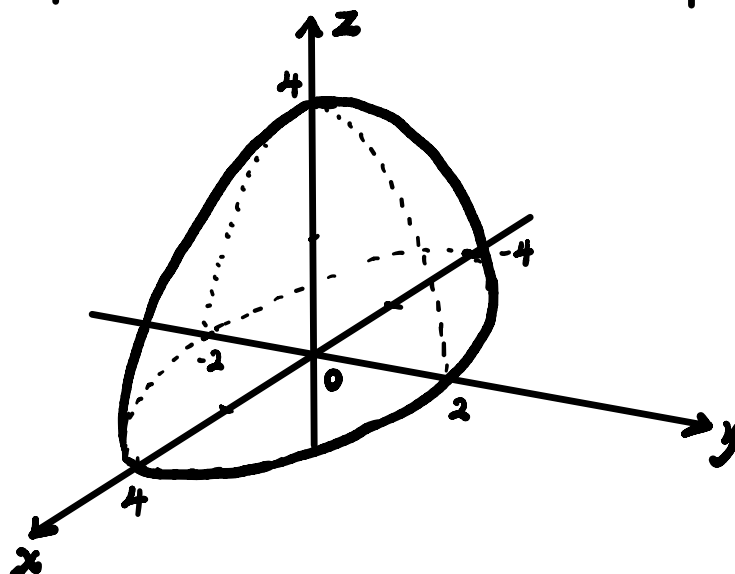
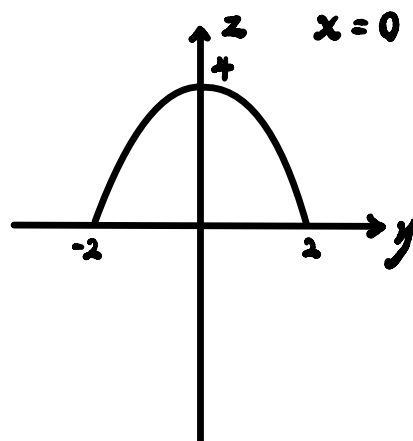
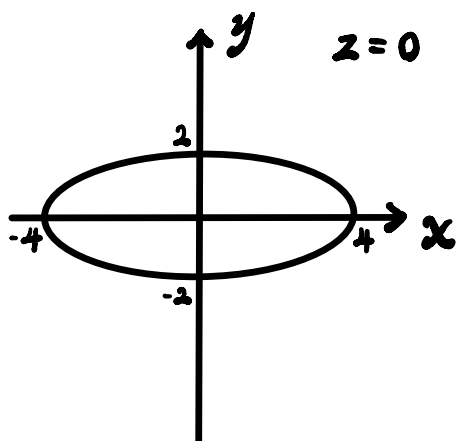
$$\therefore a \cdot b = \|a\|\|b\| \cos \theta, \quad -1 \leq \cos \theta \leq 1$$

$$\therefore |a \cdot b| \leq \|a\|\|b\| \quad \therefore a \cdot b \leq \|a\|\|b\|$$

$$\therefore (\|x\| + \|y\| + \|z\|)^2 - (\|x+y+z\|)^2 \geq 0$$

$$\therefore \|x\| + \|y\| + \|z\| \geq \|x+y+z\|$$

(c) for $z = -\frac{x^2}{4} - y^2 + 4$ ($z \geq 0$)



At $(x, y) = (1, 1)$,

$$z = z_0 + \left. \frac{dz}{dx} \right|_{x=1} (x - x_0) + \left. \frac{dz}{dy} \right|_{y=1} (y - y_0)$$

$$= \frac{11}{4} + \left(-\frac{1}{2}\right)(x-1) + (-2)(y-1)$$

$$= -\frac{1}{2}x - 2y + \frac{21}{4}$$

$$\begin{aligned}
 \text{d) } \nabla(fg) &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) (fg) \\
 &= \frac{\partial(fg)}{\partial x} + \frac{\partial(fg)}{\partial y} + \frac{\partial(fg)}{\partial z} \\
 &= f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x} + 2 \frac{\partial g}{\partial x} \frac{\partial f}{\partial x} \\
 &\quad + f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y} + 2 \frac{\partial g}{\partial y} \frac{\partial f}{\partial y} \\
 &\quad + f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z} + 2 \frac{\partial g}{\partial z} \frac{\partial f}{\partial z} \\
 &= f \nabla g + 2(\nabla f \cdot \nabla g) + g \nabla f
 \end{aligned}$$